

Jemuel Daniel

Full-Stack & Interface Engineer | AI/SI Focus

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Profile

Software engineer and founder with five years of hands-on software development. Builds interfaces and agent workflows, with particular attention to context, recoverable execution and clear failure states. Works across frontend, APIs and runtime behavior.

AI/SI means artificial intelligence and superintelligence; safe superintelligence is a long-term research interest.

Experience

Founder and software engineer

Building my own startup, with hands-on work spanning product interfaces, agent workflows, backend contracts and systems experiments. The selected projects below describe my implementation work and its current validation scope.

Selected engineering work

Recoverable agent workspace

Agent workflows, context management and execution recovery

- Implemented conversation lineage and scoped history so resumed work can recover relevant context without mixing unrelated sessions.
- Built partial-output recovery, execution deadlines and single-flight readback so an uncertain result can be checked before another operation starts.
- Added account-drift checks and explicit recovery states to keep authorization context and user-visible execution status aligned.

Implementation and regression test cases are in place; end-to-end database, provider and browser integration remains in progress.

Embedded applications and product interfaces

React, TypeScript and host SDK integration

- Built embedded mini-app interfaces with host SDK handoff and session hydration, connecting application state to the surrounding host experience.
- Implemented search, challenge flows, leaderboards and media surfaces, with responsive layouts and mobile interaction states.
- Made loading, empty, blocked and failed states explicit so users can understand what happened and what to do next.

Interface and integration paths are implemented; live end-to-end browser validation remains in progress.

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Systems correctness

Durable API operations

Operation identity and transaction recovery

- Bound durable operation IDs to the actor and request payload so retries resolve the original operation and cannot silently change its meaning.
- Implemented recovery of the original result after a simulated SQL response loss. Fixture tests exercise this recovery path; live database behavior still needs integration validation.

Experimental compiler and virtual machine

Language implementation and bounded execution

- Implemented a lexer, parser and type/effect checking, with high- and mid-level intermediate representations, a custom virtual machine and WebAssembly work.
- Added verification before execution and step/frame bounds. Developed mutation and verifier regression tests around malformed input and execution limits.

Additional engineering work

Training and evaluation infrastructure

Implemented Python orchestration, dataset admission and checkpoint lineage to connect experiment inputs, execution and resulting artifacts, with controls for repeatable evaluation.

Smart contract transaction checks

Implemented Solidity transaction checks with regression test definitions for replay, signature and transaction-state boundaries. Audit and live deployment validation are outside the scope of this work.

Core technologies

Primary TypeScript, JavaScript, Python, SQL / PostgreSQL

Interfaces React, Next.js, responsive UI, streaming state, host SDKs

Agents and APIs Context and retrieval, agent workflows, Node.js, API contracts, MCP

Systems Rust, Solidity, compiler IR, virtual machines, WebAssembly

Availability

Available to start immediately. Open to remote work and international relocation, subject to visa and work authorization.